Weekly report

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This week

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- Competition: Discussion of modeling choices

Positive advantage: a fixed-policy test

Results from *Final note: Intermediate forgetting* on Overleaf.

Let U_ρ^i be the value of always choosing product i. On the invariant limit set $\mathcal{K}_\rho$, define

$$a_\rho(z) = -\Delta c + \gamma \Delta p [V_\rho(T_\rho^s z) - V_\rho(T_\rho^f z)], \tag{1}$$

$$\mathcal{A}_\rho = \max_{z \in \mathcal{K}_\rho} a_\rho(z). \tag{2}$$

Proposition 1: Characterization of positive advantage

For $\rho \in (0,1)$,

$$\mathcal{A}_\rho > 0 \quad \Longleftrightarrow \quad \max_{z \in \mathcal{K}_\rho} [U_\rho^1(T_\rho^s z) - U_\rho^1(T_\rho^f z)] > \frac{\Delta c}{\gamma \Delta p}.$$

Persistent use of product 2

Proposition 2

Fix $\rho \in (0,1)$. If $\mathcal{A}_\rho > 0$, there exists $\eta > 0$ such that, from every initial state and under any optimal policy,

$$\liminf_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}\{i_n = 2\} \geq \eta \qquad \text{almost surely.}$$

Here i_n is the product chosen at the nth interaction with provider A.

Product 2 has a positive lower asymptotic frequency of use. In particular, the provider uses it infinitely often.

State-dependent choices and indifference

Define the fixed-policy advantages

$$a_\rho^i(z) = -\Delta c + \gamma \Delta p[U_\rho^i(T_\rho^s z) - U_\rho^i(T_\rho^f z)], \quad i = 1, 2,$$
$$\mathcal{A}_\rho^1 = \max_{z \in \mathcal{K}_\rho} a_\rho^1(z), \qquad \mathcal{A}_\rho^2 = \min_{z \in \mathcal{K}_\rho} a_\rho^2(z).$$

Lemma 1

If $\rho \in (1/2, 1)$, $\mathcal{A}_\rho^1 > 0$, and $\mathcal{A}_\rho^2 < 0$, each product is strictly optimal at some states in $\mathcal{K}_\rho$. Moreover,

$$\exists\, x^* \in (0, 1): \qquad a_\rho(x^*, 1 - x^*) = 0.$$

- Connectedness and continuity guarantee a zero; an interval of indifference is also possible.
- A threshold structure needs further restrictions on the sign pattern: the number, order, and direction of switches remain undetermined.

Persistent use of both products

Corollary 1

Fix $\rho \in (0,1)$. If $\mathcal{A}_\rho^1 > 0$ and $\mathcal{A}_\rho^2 < 0$, there exist $\eta_1, \eta_2 > 0$ such that, from every initial state and under any optimal policy,

$$\liminf_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}\{i_n = j\} \geq \eta_j \quad \text{almost surely,} \qquad j = 1, 2.$$

- Both products are used infinitely often, each with a positive lower asymptotic frequency.

Cost region with both products strictly optimal at some states

For $i \in \{1, 2\}$, let

$$H_\rho^i(z) = \mathbb{E}_z^i \left[\sum_{t=1}^{\infty} \gamma^{t-1} \mathbf{1}\{y_t = A\} \right]$$

be the expected discounted number of purchases from seller A when product i is always used. Here t counts calendar periods, and $\mathbb{E}_z^i$ denotes expectation starting from z under this fixed policy.

Proposition 3

Fix $\rho \in (1/2, 1)$, $\gamma \in (0, 1)$, $p_0 \in (0, 1)$, $0 < p_1 < p_2 < 1$, and $0 \leq c_1 < R$. Define

$$\underline{c}_2 = \frac{c_1 + \gamma \Delta p \, R \min\limits_{z \in \mathcal{K}_\rho} \left[H_\rho^2(T_\rho^s z) - H_\rho^2(T_\rho^f z) \right]}{1 + \gamma \Delta p \min\limits_{z \in \mathcal{K}_\rho} \left[H_\rho^2(T_\rho^s z) - H_\rho^2(T_\rho^f z) \right]},$$

$$\overline{c}_2 = c_1 + \gamma \Delta p \, (R - c_1) \max_{z \in \mathcal{K}_\rho} \left[H_\rho^1(T_\rho^s z) - H_\rho^1(T_\rho^f z) \right].$$

These bounds do not depend on c_2. For any $c_2 \in (c_1, R)$,

$$\mathcal{A}_\rho^1 > 0 \text{ and } \mathcal{A}_\rho^2 < 0 \quad \Longleftrightarrow \quad \underline{c}_2 < c_2 < \overline{c}_2.$$

Thus $(\underline{c}_2, \overline{c}_2) \cap (c_1, R)$ is exactly the cost region where each product is strictly optimal at some state in $\mathcal{K}_\rho$. For every cost in this region, there exists $x^* \in (0, 1)$ such that $a_\rho(x^*, 1 - x^*) = 0$.

Fluid model: the transformed payoff

Proposition 4: Shape of the transformed payoff

Assume $D'(x) > 0$ and D' is log-concave on $[0, 1]$. Then the set $\{x \in [0, 1] : F'(x) > 0\}$ is an interval, possibly empty or meeting a boundary. If nonempty, F decreases before this interval, increases within it, and decreases after it, with absent segments omitted. If it is empty, product 1 is optimal everywhere.

Fluid model: a single interval of product 2

Theorem 1

Suppose, for some $0 \leq a < m \leq 1$,

$$F' < 0 \text{ on } (0, a) \cup (m, 1), \qquad F' > 0 \text{ on } (a, m),$$

and every interior zero of F' is simple.

An optimal policy can be selected that uses product 2 on at most one interval and product 1 outside it. As x increases, the possible nonconstant patterns are

$$1 \to 2, \qquad 2 \to 1, \qquad 1 \to 2 \to 1.$$

If F is nonincreasing, product 1 is optimal everywhere; if F is nondecreasing, product 2 is optimal everywhere.

Existence of a nonconstant threshold policy

Corollary 2

Under the assumptions of Theorem 1, if in addition we have

$$\max_{x\in[0,1]}\{-\Delta c+\lambda\Delta p(U_\rho^1)'(x)\}>0,$$

$$\min_{x\in[0,1]}\{-\Delta c+\lambda\Delta p(U_\rho^2)'(x)\}<0.$$

then there is an optimal policy with either one or two interior thresholds, with one of the three patterns in Theorem 1.

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- How much information we want the sellers to know?
- Go to Francisco's proposal